

Data visualization and modelling

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3-Parametric Bootstrap

Overview

- 1- The method
- 2- Examples of application

The name "bootstrap" refers to the analogy with pulling oneself up by one's own bootstraps.

Introduced by Bradley Efron in 1979,

- B. Efron. Bootstrap methods: another look at the jackknife. *Annals of Statistics*, 7:1-26, 1979.
- B. Efron. Nonparametric estimates of standard error: the jackknife, the bootstrap, and other methods. *Biometrika*, 68: 589-599, 1981.



THE PARAMETRIC BOOTSTRAP APPROACH

- Assume data come from random variables from a particular distribution (E.g., log-normal, Poisson, Normal, etc)
- Use data to *estimate* the parameters of the distribution (E.g., mean, variance, etc)
- Use a random number generator to create samples from the pattern distribution with the same size as original.
- Calculate the sample mean or the statistics of interest. Confidence intervals can be calculated from the quantiles.

THE PARAMETRIC BOOTSTRAP APPROACH

Resampling methods: a practical guide to data Analysis. Phillip I. Good (2006). Chapter 2, p.21.

2.3.5 Parametric Bootstrap

When we know the form of the population distribution, the use of the *parametric bootstrap* to obtain interval estimates may prove advantageous either because the parametric bootstrap provides more accurate answers than textbook formulas or because no textbook formulas exist. The parametric bootstrap is particularly valuable when the form of the distribution is known and confidence intervals are required for statistics such as the variance or the 90th percentile that depend heavily on values in the tails.

Suppose we know the observations come from a normal distribution and want an interval estimate for the 95th percentile. We would draw repeated bootstrap samples from the original sample, use the mean and variance of the bootstrap sample to estimate the parameters of a normal distribution, and then derive a bootstrap estimate of the 95th percentile from tables of an $N(0, 1)$ distribution.

A simple example: CI for a proportion.

In a sample of 15 individuals we find 3 of them having certain sickness. What is the prevalence?

- The estimated prevalence is $3/15=0.2$
- We simulate samples of size 15 coming from a Bernoulli distribution with $p=3/15$, and we count the number of successes. This is equivalent to simulate binomials with $n=15$ and $p=3/15$.
- For each sample we calculate the prevalence.

```
pr<-rbinom(1000,15,3/15)/15
```

```
0.26666667, 0.13333333, 0.20000000, 0.06666667, 0.00000000, 0.13333333, ...
```

- Bootstrap estimate:

$\text{mean}(\text{pr})$ 0.19

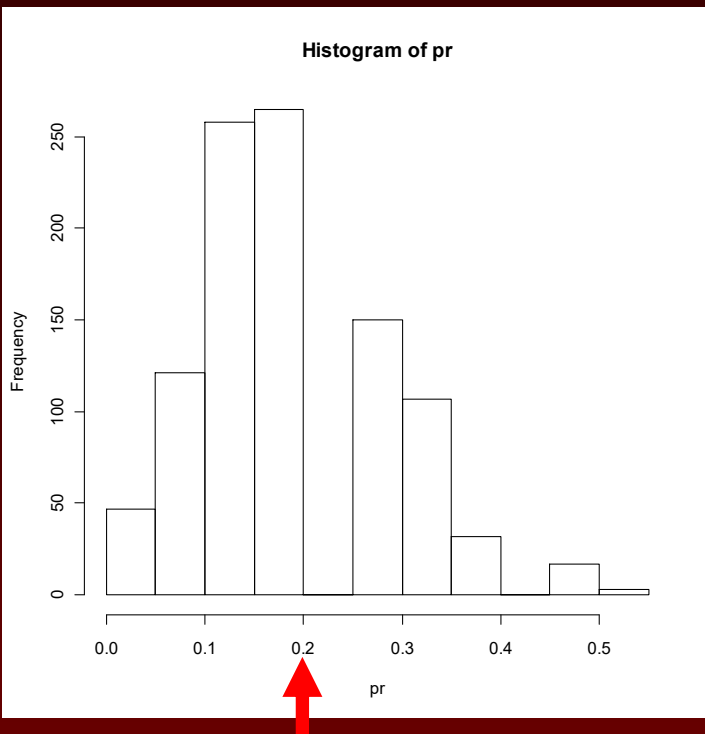
- Expected Bias:

$\text{mean}(\text{pr}) - 3/15$ -0.01

- Estimated standard error:

$\text{SD}(\text{pr})$ 0.1026

- *So prevalence estimate is: 0.19 (SE 0.103)*
(using the classical formula the SE is 0.103)



We can calculate (95%) confidence intervals using the bootstrap SE:

0.19 ± 0.202 (This is not a good CI !)

We can also calculate (95%) confidence intervals using the quantiles of the bootstrap replicates:

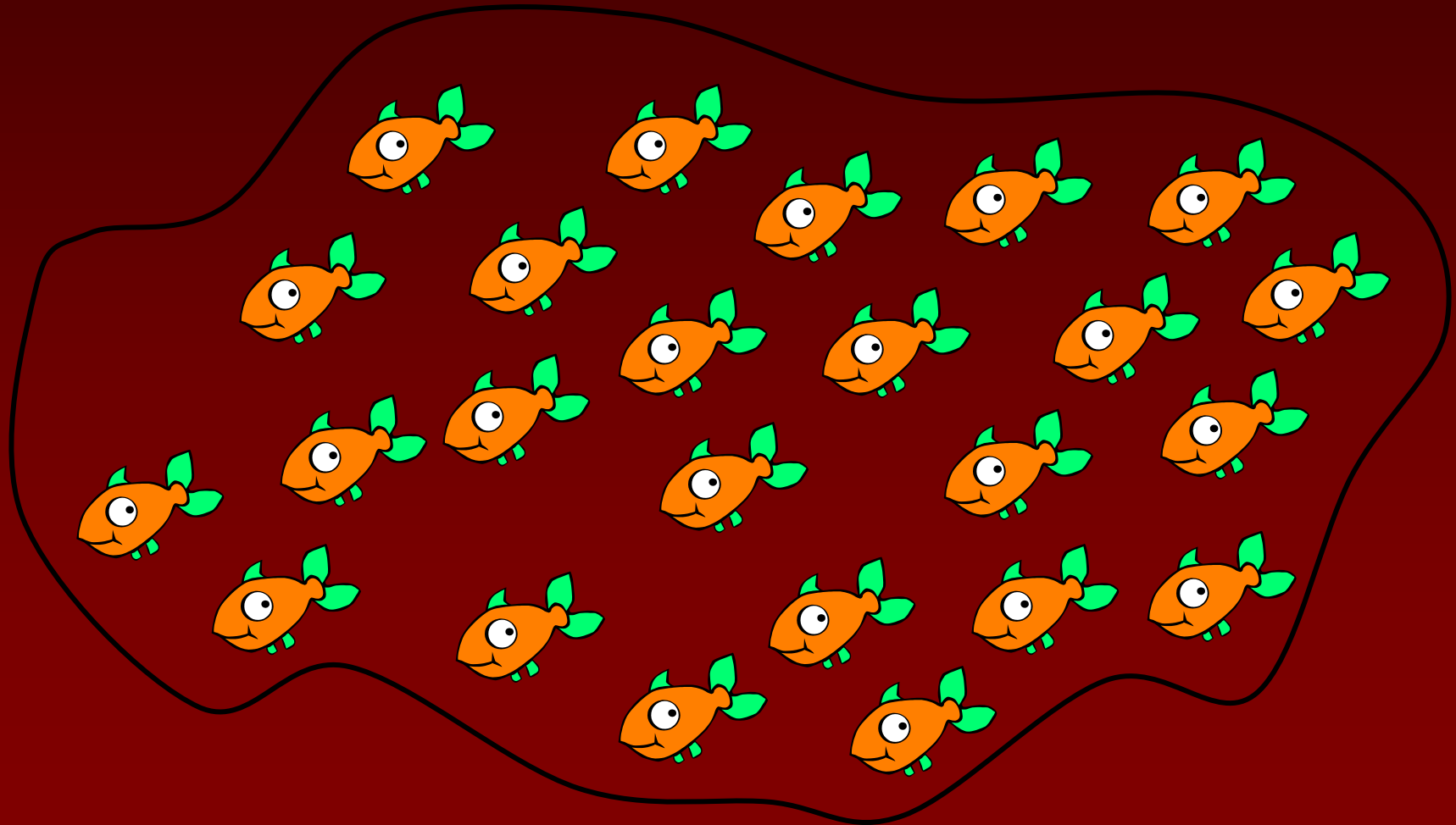
```
quantile(pr, probs = c(0.025, 0.975))
```

2.5% 97.5%

0.0 0.4

This procedure is much better as it leads to the interval $[0, 0.4] : 0.2 \pm 0.2$

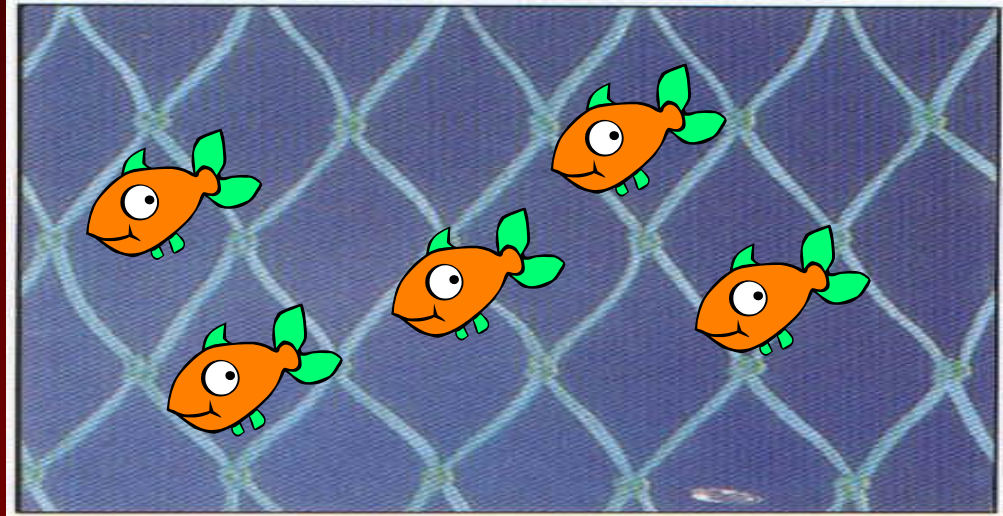
How many fishes are in the lake?



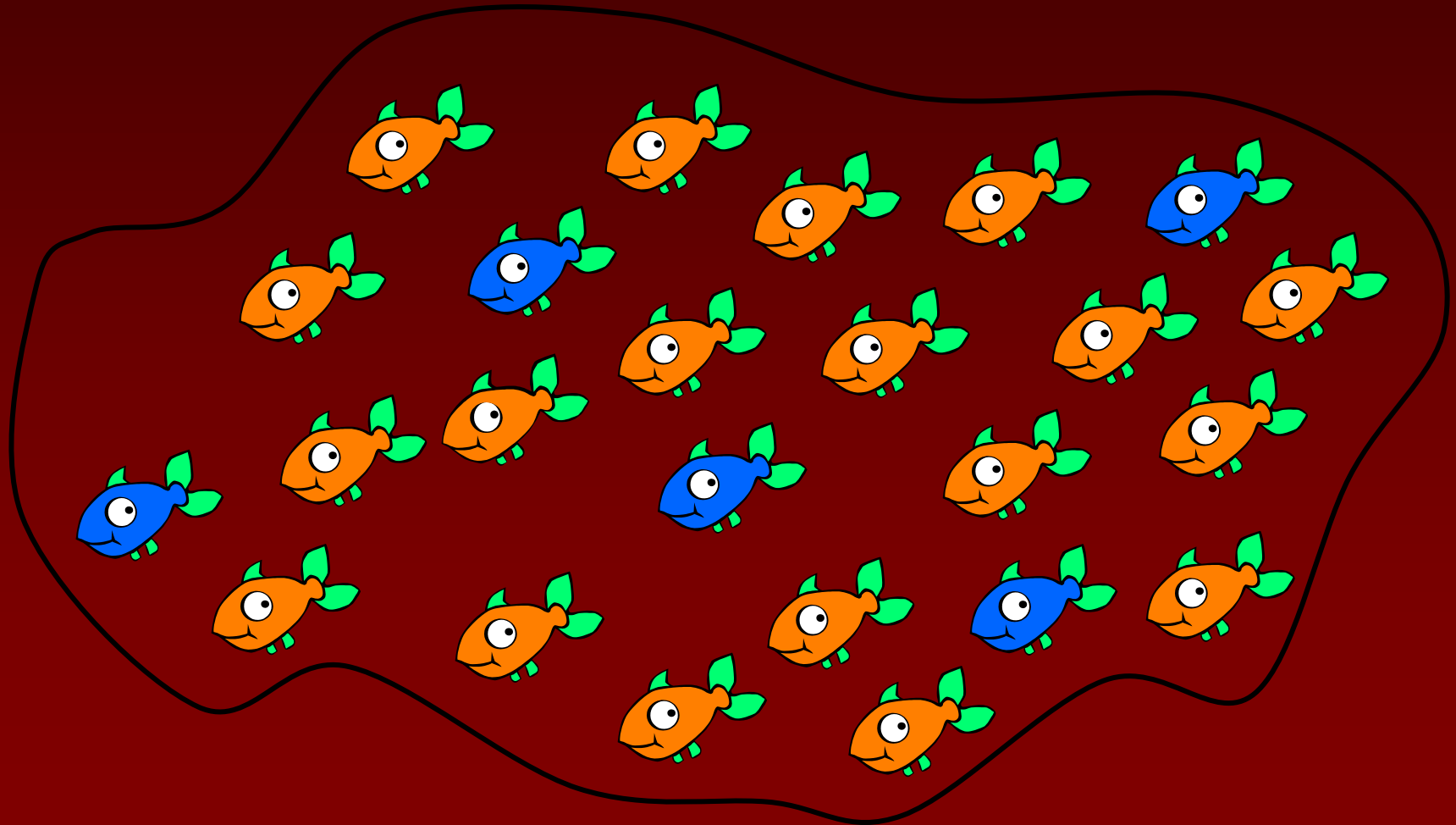
A Capture-Recapture tutorial

- The researcher uses traps to capture a group of individuals alive. Each of these individuals is marked with a unique identifier and then is released unharmed back into the environment.
- Sufficient time is allowed to pass for the marked individuals to redistribute themselves among the unmarked population. Next, the researcher returns and captures another sample of individuals.
- Some of the individuals in this second sample will have been marked during the initial visit and are now known as recaptures. Other animals captured during the second visit will not have been captured during the first visit to the study area. This information allows to estimate the abundance of the population.

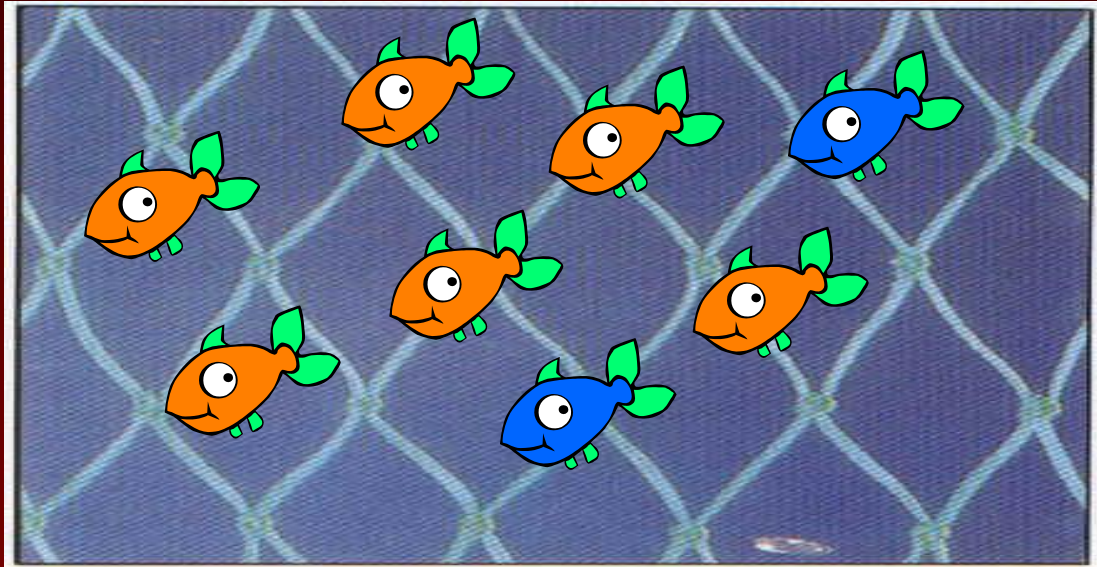
We capture a group of individuals alive and they are marked.



Sometime after, they are redistributed among the unmarked population.



We capture another sample of individuals



For instance, we have captured 8 and 2 of them are marked.

Let n_1 be the number of individuals captured and marked on the first round, n_2 the number of individuals captured on the second round, m the number of individuals that are found marked on the second round, and N is the total population size. It is straightforward to see that,

$$\frac{n_1}{N} \approx \frac{m}{n_2}$$

and consequently

$$N \approx \frac{n_1 n_2}{m}$$

$$\hat{N} = \frac{n_1 n_2}{m}$$

This is known as the Lincoln-Petersen estimator of the population size.

For our example,

$$n_1 = 5$$

$$n_2 = 8$$

$$m = 2$$

$$\hat{N} = \frac{5 \times 8}{2} = 20$$

<https://www.youtube.com/watch?v=Hi460--IUhM&t=3s>

- It has no sense when $m=0$.
- This estimator has a bias that can be corrected using the Chapman (1951) estimator:

$$\hat{N}_c = \frac{(n_1 + 1)(n_2 + 1)}{(m + 1)} - 1$$

For our example:

$$\hat{N}_c = \frac{(5 + 1)(8 + 1)}{(2 + 1)} - 1 = 17$$

The study of these estimators are based on the fact that, fixed n_1 and n_2 , the random variable m follows an hypergeometric distribution. That is,

$$P(m = k) = \frac{\binom{n_1}{k} \binom{N - n_1}{n_2 - k}}{\binom{N}{n_2}}$$

The values of m are in the interval,

$$\left[\max(0, n_2 + n_1 - N), \min(n_1, n_2) \right]$$

The variance of the estimator can be calculated by using the expression,

$$V(\hat{N}_c) = \frac{(\mathbf{n}_1 + 1)(\mathbf{n}_2 + 1)(\mathbf{n}_1 - m)(\mathbf{n}_2 - m)}{(m + 2)(m + 1)^2}$$

It allows to calculate approximated confidence intervals (95%):

$$N = \hat{N}_c \pm 1.96 \sqrt{V(\hat{N}_c)}$$

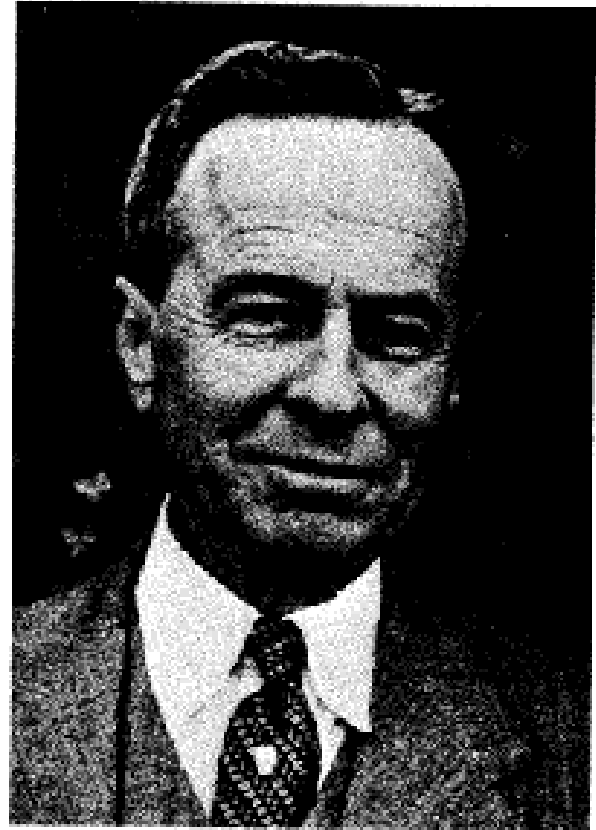
For our example,

$$N = 17 \pm 1.96 \sqrt{27} = 17 \pm 10$$

The “popularizers” of the method



Carl George Johannes Petersen



Frederick C. Lincoln

Model assumptions

- The population is closed.
- Both samples are random samples.
- All the individuals have the same probability to be captured.
- Capture and marking do not change the probability, that is, all individuals have the same probability of being captured in the second sample.
- No marks fall off individuals between captures.

The precursor



The first use of the capture-recapture technique can be traced back to Laplace, who used it to estimate the population size of France in 1786.

An incomplete census was the first capture (n_1). The individuals recorded in a survey were the second capture (n_2). The number of individuals that were coincident in both lists was m .

Classical example in Bangkok (AJPH, 1994)

Estimating the Number of HIV-Infected Injection Drug Users in Bangkok: A Capture–Recapture Method

Timothy D. Mastro, MD, Dwip Kitayaporn, MD, DrPH, Bruce G. Weniger, MD, MPH, Suphak Vanichseni, MD, MPH, PhD, Vuti Laosunthorn, PhD, Thongchai Uneklabh, MD, Chintra Uneklabh, MD, Kachit Choopanya, MD, MPH, and Khanchit Limpakarnjanarat, MD, MPH

TABLE 3—Estimates of the Number of Opiate Users in Bangkok (n), by Unstratified and Age-Stratified Analyses

Age Group	Capture 1	Capture 2	No. of Matches	n	95% Confidence Interval
Unstratified analysis					
All	4064	1540	171	36 600	31 538, 41 662
Stratified analysis					
< 20 y	76	78	4	1482	105, 2859
20–29 y	1378	699	76	12 674	10 059, 15 289
30–39 y	2106	644	80	16 953	13 543, 20 363
> 39 y	491	118	11	5267	2336, 8198
All	4051	1539	171	36 376	30 996, 41 757

THE INTERNATIONAL JOURNAL *of* SCIENCE IN SOCIETY

Using Capture-recapture to Estimate the Prevalence of Intimate Partner Violence: The Gender Symmetry Debate

Lori A. Post, Yale University, CT, USA

Nancy J. Mezey, Monmouth University, New Jersey, USA

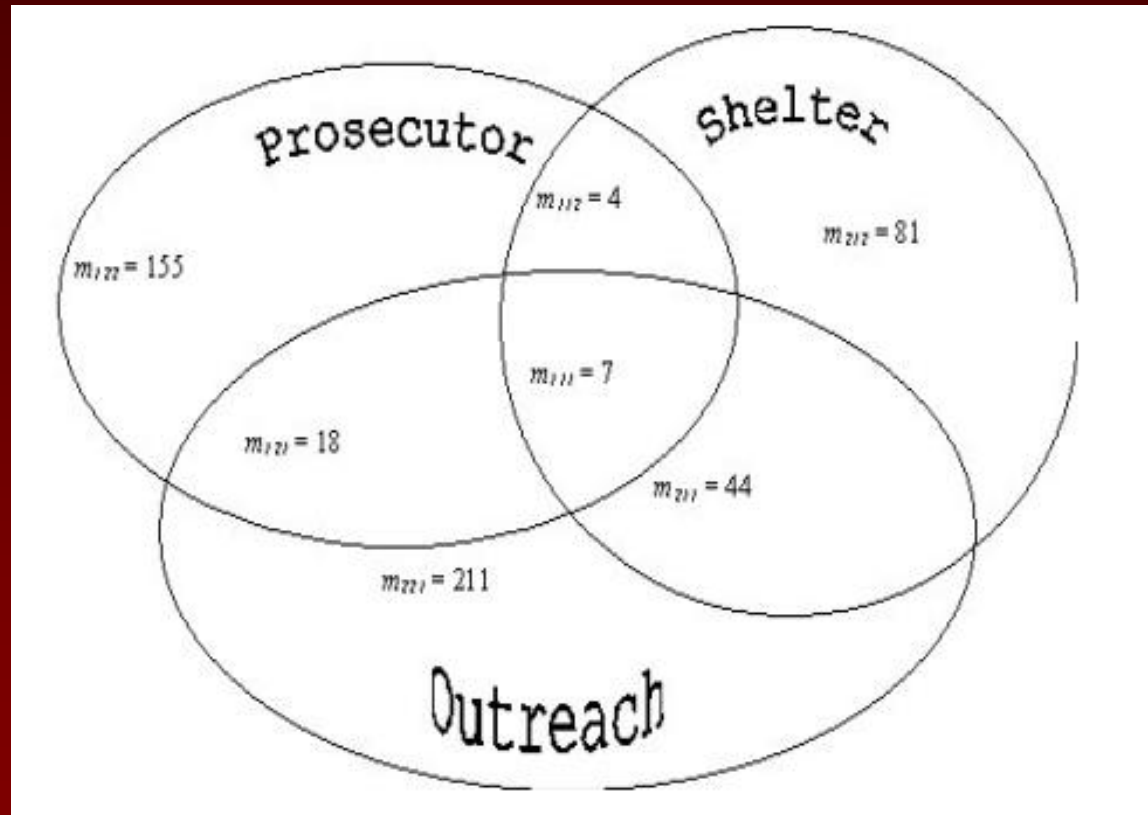
Christopher D. Maxwell, Michigan State University, MI, USA

Karin Rhodes, University of Pennsylvania School of Medicine, PA, USA

Using Capture-recapture to Estimate the
Prevalence of Intimate Partner Violence: The
Gender Symmetry Debate

The authors consider three lists:

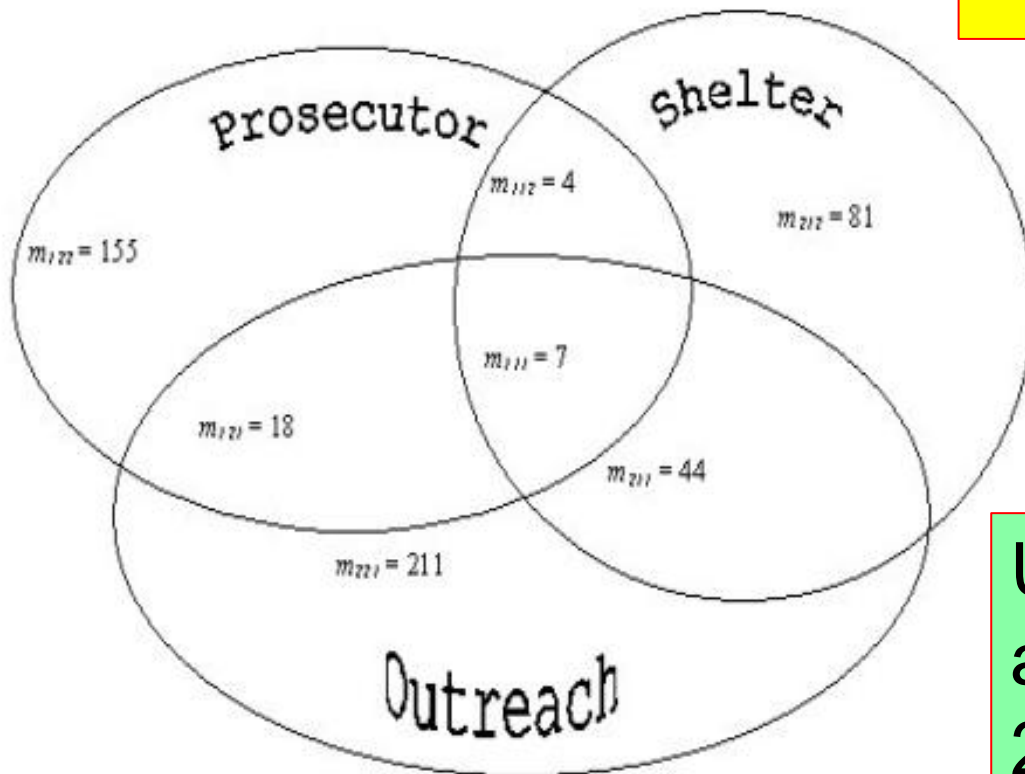
- Prosecutor's office
- Local domestic violence shelter
- Family Outreach agencies



We consider only two lists:

- Prosecutor = $155+4+18+7 = 184$
- Outreach = $211+44+7+18 = 280$
- Both lists = $18+7 = 25$

$$\hat{N} = \frac{184 \times 280}{25} \sim 2061$$



Using the three lists the authors find an estimate of 2337 women.

Parametric Bootstrap—an example

Mark-Recapture Estimate

Mark 184 women
Recapture 280
of which 25 Marked

What is population size?

Chapman estimate is 1998

What is confidence in this estimate, expected bias?

- Parametric Bootstrap Replicates:
 - 1998 women, mark 184, recapture 280
How many marked?
 - From simulation (**m**), using
 $x \leftarrow \text{rhyper}(10000, 184, 1814, 280)$
23, 23, 26, 28, 26, 30, 22, 25, 21, 30, 30,...
 - Calculate population estimates,
 $n \leftarrow (185 * 281 / (x + 1)) - 1$
2165.042, 2165.042, 1924.370, 1791.586,...

- Bootstrap estimate:

$\text{mean}(n)$ 1993.324

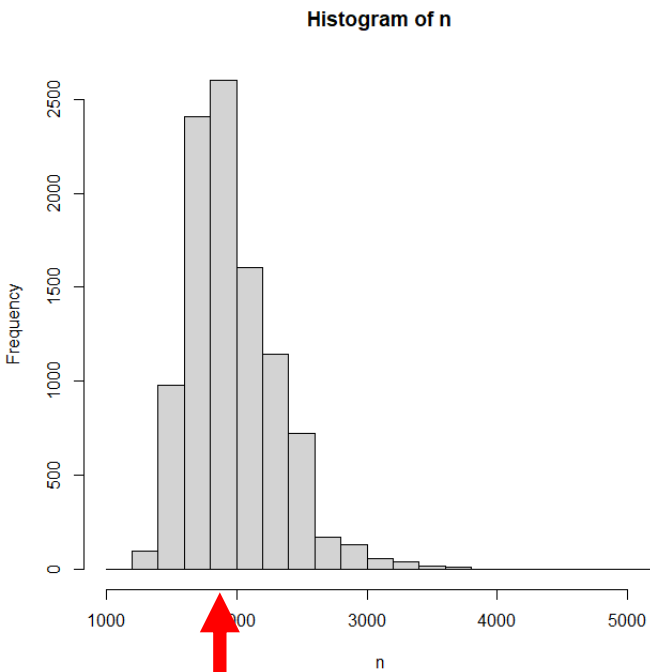
- Expected Bias:

$\text{mean}(n) - 1998$ -4.676

- Estimated standard error:

$\text{SD}(n)$ 356.6

- *So population estimate is: 1993 (SE 356.6)*



We can calculate (95%) confidence intervals using the bootstrap SE:

$$1993 \pm 699$$

We can also calculate (95%) confidence intervals using the quantiles of the bootstrap replicates:

```
quantile(n, probs = c(0.025, 0.975))
```

2.5% 97.5%

1443.028 2887.056

This procedure is much better and it leads to the interval,

$$[1443, 2887] : 2165 \pm 722$$

THE PARAMETRIC BOOTSTRAP IN REGRESSION

There are two types of observations for which we can apply linear least squares regression:

A. We are making random observations of X and Y together.

B. We are testing at different specific values of X to determine the response in Y .

For type A, both X and Y are Bootstrapped together in pairs and calculate the regression coefficients with each Bootstrap replicate.

THE PARAMETRIC BOOTSTRAP IN REGRESSION

For type B, we need to retain the selected nature of the X variable. The random element is now just the response of the Y variable to the X value and it is this random component that should be bootstrapped: the X values are fixed since they were predetermined rather than resulting from a random sample from a distribution. Assuming that the random variations about the regression line are homoscedastic and that the straight line relationship is correct, the only random variable involved is that producing the variations about the line and so we Bootstrap the residuals about that line.

An example: Correlation between gill weight and body weight in 12 crabs

Gill (mg)	Body (g)
-----------	----------

1590	14,4
------	------

1790	15,2
------	------

10	11,3
----	------

450	2,5
-----	-----

3840	22,7
------	------

23	14,9
----	------

10	1,41
----	------

32	15,81
----	-------

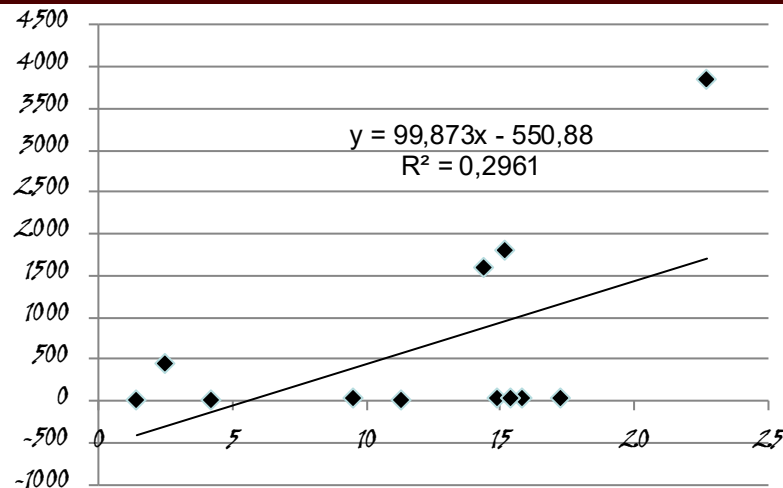
8	4,19
---	------

22	15,39
----	-------

32	17,25
----	-------

21	9,52
----	------

$r = 0,54417901$



It corresponds to a type A regression model and we can assume that the observations come from a bivariate normal distribution.

- First of all we have to estimate the 5 parameters:

```
gill<-c(1590,1790,10,450,3840,23,10,32,8,22,32,21)
```

```
body<-c(14.40,15.20,11.30,2.50,22.70,14.90,1.41,15.81,  
4.19,15.39,17.25,9.52)
```

```
mg<-mean(gill);mb<-mean(body)
```

```
sdg<-sd(gill);sdb<-sd(body); rho<-cor(gill,body)
```

We need to simulate bivariate normal samples:

Bivariate Normal - Windows Internet Explorer

Escriba lo que desea buscar en Web (Ctrl+B)

http://www.statisticalengineering.com/bivariate_normal.htm

Favoritos Bivariate Normal

1) Generate two, uncorrelated, standard normal variates, z_1 and z_2 .

$$z \sim \Phi(\mu = 0, \sigma^2 = 1)$$

2) Compute the *correlated* X_1 and X_2

$$X_1 = \mu_1 + \sigma_1 z_1$$
$$X_2 = \mu_2 + \sigma_2 \left[z_1 \rho + z_2 \sqrt{1 - \rho^2} \right]$$

3) X_1 and X_2 will have means μ_1 and μ_2 , standard deviations σ_1 and σ_2 , and correlation ρ .

Cautions:

1) While it is almost always possible to calculate means and standard deviations, that doesn't mean the data have a normal distribution.

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- Generate the parametric bootstrap replicates and calculate the statistics of interest.

```
nsim<-1000 ; n<-length(gill)
slope<-numeric(nsim);interc<-numeric(nsim)
r<-numeric(nsim)
for(i in 1:nsim){
  rgill<-numeric(n);rbody<-numeric(n)
  for(j in 1:n){
    z1<-rnorm(1,0,1);z2<-rnorm(1,0,1)
    rgill[j]<-mg+sdg*z1 ; rbody[j]<- mb+sdb*(z1*rho+z2*sqrt(1-
    rho^2))}
  a<-lm(rgill~rbody)
  slope[i]<-a$coefficient[2] ; interc[i]<-a$coefficient[1]
  r[i]<-cor(rgill,rbody)}
```

- Calculate confidence intervals.

`quantile(slope, probs = c(0.025, 0.975))`

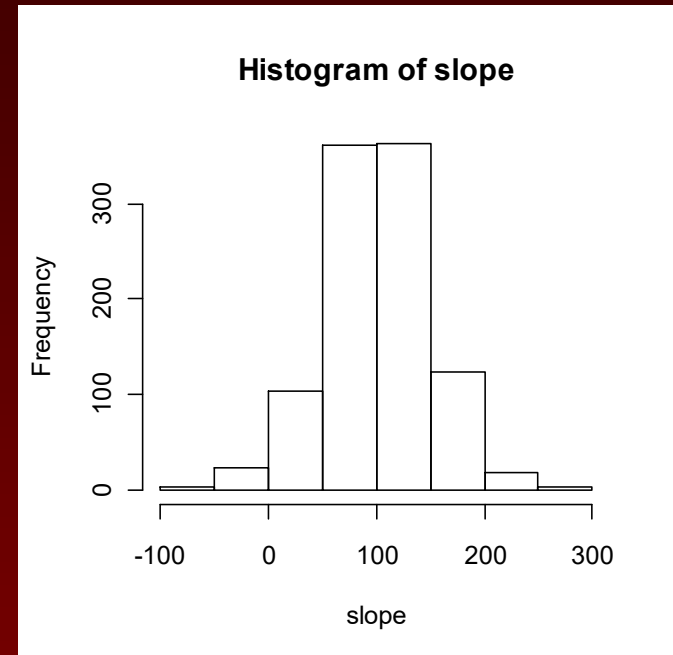
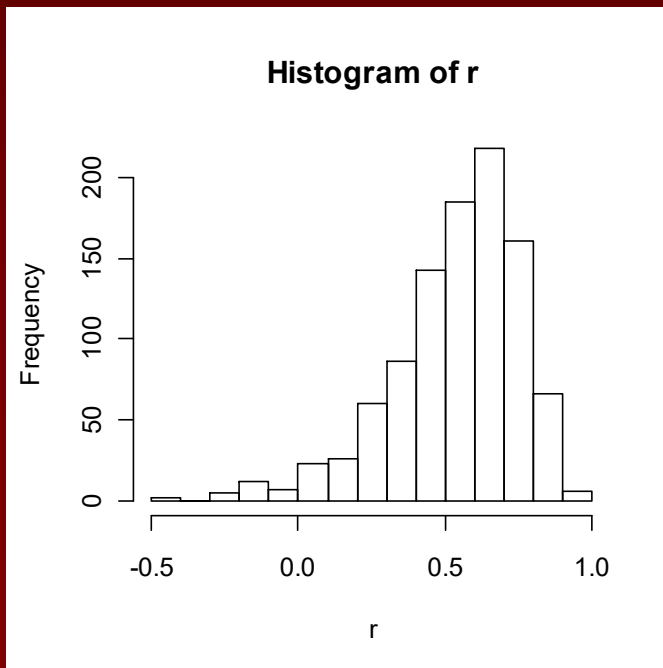
2.5% 97.5%

-6.045932 198.526366

`quantile(r, probs = c(0.025, 0.975))`

2.5% 97.5%

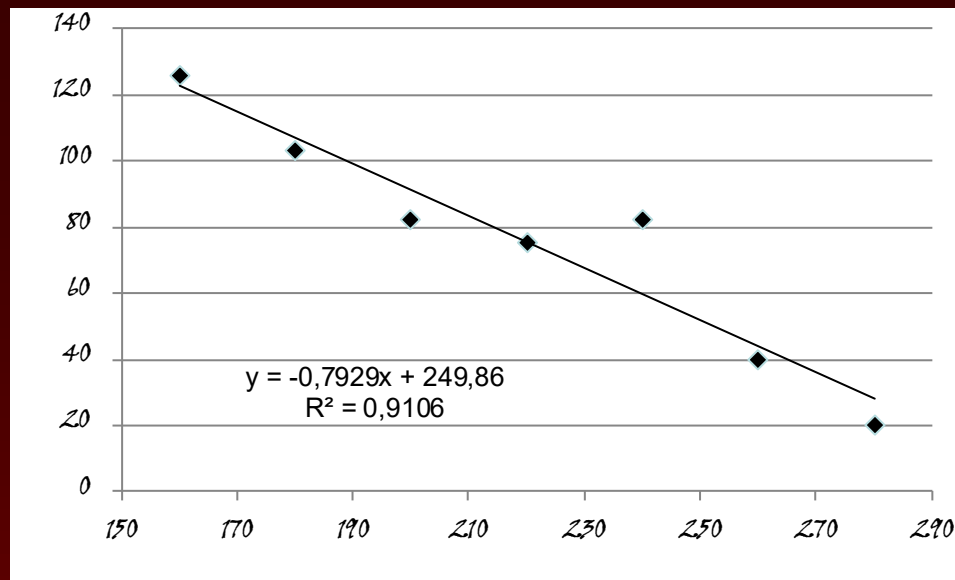
-0.02811273 0.84489811



An example: Suppose that your market research of real estate investments reveals the following sales figures for new homes of different prices over the past year.

Price (Thousands of \$):	160	180	200	220	240	260	280
Sales of New Homes:	126	103	82	75	82	40	20

We would like to use these data to construct a demand function for the real estate market. (A demand function gives demand y , measured here by annual sales, as a function of unit price, x .)



It corresponds to a type B regression model and we can assume that the errors come from a normal distribution: $y_i = a + bx_i + \varepsilon_i$

- First of all we have to estimate the parameters, a, b and the standard deviation of the errors:

```
price<-c(160, 180, 200, 220, 240, 260, 280 )
```

```
sales<-c(126, 103, 82, 75, 82, 40, 20 )
```

```
a<-lm(sales ~ price); b<-summary(a)
```

```
b
```

Residuals:

1	2	3	4	5	6	7
3.0000	-4.1429	-9.2857	-0.4286	22.4286	-3.7143	-7.8571

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	249.8571	24.8413	10.058	0.000166 ***
price	-0.7929	0.1111	-7.137	0.000838 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.76 on 5 degrees of freedom

Multiple R-squared: 0.9106, Adjusted R-squared: 0.8927

F-statistic: 50.93 on 1 and 5 DF, p-value: 0.0008385

- Generate the parametric bootstrap replicates and calculate the statistics of interest.

```
nsim<-1000 ; n<-length(price)
slope<-numeric(nsim); interc<-numeric(nsim)
r<-numeric(nsim); rsales<-numeric(n)
for(i in 1:nsim){
  error<-rnorm(n,0,b$sigma)
  rsales<-a$coefficient[1]+a$coefficient[2]*price+error
  c<-lm(rsales~price)
  slope[i]<-c$coefficient[2] ; interc[i]<-c$coefficient[1]
  r[i]<-cor(rsales,price)}
```

- Calculate confidence intervals.

`quantile(slope, probs = c(0.025, 0.975))`

2.5% 97.5%

-1.0079755 -0.5773406

`quantile(r, probs = c(0.025, 0.975))`

2.5% 97.5%

-0.9929142 -0.8802001

